PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Bus Route Optimizer

Team Name: *Team #*	*Bus Optimizers*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Kant, Krishna – 2510380211* *2510380211@geu.ac.in* 
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Team member 3 *(Last Name, name: student ID: email, picture):*	*Singh, Aditi – 2510012426* *2510012426@geu.ac.in* 
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

The Bus Route Management System is a C++ project designed to make bus booking and route management easier and more efficient. It uses data structures and algorithms such as graphs, BFS, and Dijkstra's algorithm to find routes and calculate the shortest paths. The system also supports ticket booking and cancellation, seat management, waiting lists, and file-based data storage. It provides separate menus for administrators and passengers, making the system simple to use and manage.

Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

The system is organized into modules for users, file handling, graphs, and bus management. BFS finds routes with the fewest stops, while Dijkstra's algorithm finds the shortest path by distance. The system also handles bus management, ticket booking, cancellations, and waiting lists. FileManager saves and loads data using text files. The project is built in C++ with a simple console-based interface.

Tasks Completed (7 pts)

The main Phase-II tasks have been completed across the team. The work included building the C++ data models, implementing the graph and routing algorithms, connecting the admin and passenger menus, adding booking and cancellation logic, and adding file-based storage.

Task Completed	Team Member
Graph implementation using adjacency lists, BFS and Dijkstra routing; system integration and overall coordination.	Krishna Kant
Bus stop and route management; implementation of bus search, route creation, and network display features.	Agrima Mishra
Ticket booking and cancellation; seat availability, fare calculation, and waiting-list queue management.	Aditi Singh
File handling and data persistence; testing system features, checking edge cases, and preparing project documentation.	Akhil Gahlawat

Challenges/Roadblocks (7 pts)

Describe the main implementation challenges and the steps being taken to solve them.

One challenge was keeping the graph consistent with the bus-stop data. The implementation handles this by mapping stop IDs to internal graph indices. Another challenge was connecting BFS and Dijkstra's algorithm with the user menus while keeping the code understandable. The routing functions remain inside the Graph class and are accessed through BusSystem.

File handling also required care because several types of records have to be saved and loaded. The project uses separate text files and toString()/fromString() functions for this. Booking adds another case because seats can become full, so a queue is maintained for waiting passengers. Further testing will focus on invalid stop names, unavailable buses, full buses, cancellation, waiting-list processing, and routes where no path exists.

Tasks Pending (7 pts)

The remaining work is mainly validation and presentation preparation. We need to test more edge cases, check data persistence, verify waiting-list behaviour after cancellation, clean up user input handling, and prepare screenshots and

documentation for the final phase.

Task Pending	Team Member (to complete the task)
Final testing, edge-case handling, screenshots and documentation	All team members

Project Outcome/Deliverables (2 pts)

(Describe what are the key outcomes / deliverables of the project. Max 200 words).

The Phase-II deliverable is a working C++ console prototype. It includes data models for buses, stops, routes, passengers, and bookings; Admin and Passenger menus; graph-based route finding; BFS for minimum-stop routes; Dijkstra's algorithm for shortest-distance routes; ticket booking and cancellation; waiting-list support; and text-file data storage. Sample bus stops, buses, routes, and road connections are also included so the system can be demonstrated directly.

Progress Overview (2 pts)

(Summarize how much of the project is done, what's behind schedule, what's ahead of schedule. Max 200 words.)

Most of the core Phase-II implementation is complete. TThe main features, including route finding, ticket booking, cancellation, seat management, and waiting lists, have been implemented. The code compiles successfully using C++17 and the application opens correctly. Further testing is underway to check errors and ensure data is saved properly. Documentation and the final presentation will be completed after testing.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

The Phase-II implementation is contained in the submitted C++ source file, full extunderscore code.cpp. The code is organized into logical sections for Utils, FileManager, Models, Graph, Users, BusSystem, and main. The most important parts are the Graph class for BFS and Dijkstra routing, BusSystem for integrating the project features, and the User/Admin/Passenger classes for the console interface.
Repository link: To be added by the team. Branch and important commit details: To be updated when the team repository is finalized.

Testing and Validation Status (2 pts)

The code has been compiled with C++17 and the application has been started successfully. The remaining feature-level tests will be documented as they are completed.

Test Type	Status (Pass/Fail)	Notes
C++ compilation and startup	Pass	Compiled with C++17 and reached the main menu successfully.

Deliverables Progress (2 pts)

The code was compiled successfully using C++17, and the application displayed the welcome screen and main menu correctly. Further testing is underway for route finding, bus search, ticket booking and cancellation, waiting lists, and data saving. The results will be recorded before the final phase.